

Solution:

Polynomial running time. Sorting the n rectangles by non-increasing height takes $O(n \log n)$ time. Then the single pass that scans through the list, forming each shelf by accumulating widths until they exceed 1, takes $O(n)$ time. Hence the overall running time is $O(n \log n)$.

Approximation ratio. Let H_G be the total height of the packing produced by the greedy algorithm, and let H^* be the height of an optimal packing.

Label the shelves produced by the algorithm $m = 1, 2, \dots, M$, and for each m let k_{m+1} be the index of the first rectangle not placed on shelf m , so by construction

$$w_{k_m} + w_{k_m+1} + \dots + w_{k_{m+1}-1} \leq 1 \quad \text{but} \quad w_{k_m} + w_{k_m+1} + \dots + w_{k_{m+1}} > 1.$$

Each shelf m is given height h_{k_m} , so

$$H_G = \sum_{m=1}^M h_{k_m}.$$

For each shelf m ,

$$\sum_{i=k_m}^{k_{m+1}} a_i = \sum_{i=k_m}^{k_{m+1}} h_i w_i \geq h_{k_{m+1}} (w_{k_m} + \dots + w_{k_{m+1}}) > h_{k_{m+1}}.$$

Summing this strict inequality over $m = 1, 2, \dots, M$ gives

$$\sum_{m=1}^M \sum_{i=k_m}^{k_{m+1}} a_i > \sum_{m=1}^M h_{k_{m+1}} = \sum_{m=2}^{M+1} h_{k_m} = \sum_{m=2}^M h_{k_m}.$$

(Since $k_{M+1} = n + 1$ and we may set $h_{n+1} = 0$.) The left-hand side is

$$\sum_{m=1}^M \sum_{i=k_m}^{k_{m+1}} a_i = \sum_{m=1}^M \left(\sum_{i=k_m}^{k_{m+1}-1} a_i + a_{k_{m+1}} \right) = A + \sum_{m=1}^M a_{k_{m+1}} < 2A$$

and in any packing into a width-1 strip of height H^* the total area satisfies $A \leq H^*$. Hence

$$\sum_{m=2}^M h_{k_m} < 2A \leq 2H^*.$$

Therefore

$$H_G = \sum_{m=1}^M h_{k_m} = h_{k_1} + \sum_{m=2}^M h_{k_m} < h_{k_1} + 2H^*.$$

Finally, since the tallest rectangle has height h_{k_1} and must fit into any feasible packing, we have

$$h_{k_1} \leq H^*,$$

and thus

$$H_G < h_{k_1} + H^* \leq 3H^*.$$

The approximation factor is at most 3.